



## United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Class Test-I Year: 2021 Trimester: Summer Program: BSCSE

Course: CSE 2215/CSI 217 Data Structure and Algorithms, Marks: 20

There are Four questions. Answer all of them. Please submit assignment by 25/07/2021 Sunday within 11:59pm(night) to lms.

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1. Show the mechanism of descending Order Quick Sort algorithm on the following data. [5]  
Here DD means last two digits of your student ID.  
10 8 20 DD 30
2. Show how the merge operation works in MergeSort(A, n) algorithm to find a [5]  
ascending order list from two separate ascending order list. Assume x is last two digits  
of your Student ID. Both ascending order lists before merging contains one number  
that is x.
3. Discuss the time complexity of counting sort. [5]
4. How many times each of the following statements will be executed? [5]

```
x= LAST_TWO_DIGITS_OF_YOUR_STUDENT_ID;  
y=x*2;  
for(i=x; i<=y; ++i){  
    sum=0;  
    for(j=1; j<=i; ++j){  
        sum=sum+i+j;  
    }  
    printf("%d ", sum);  
}
```